

Challenge (Habanero)

Q1. In a gaming leaderboard, the slope of the line representing the score increase is 3. What is the slope of a line perpendicular to it?

(A) $-\frac{1}{3}$

(B) $\frac{1}{3}$

(C) 3

(D) -3

(E) 0

Q2. A coder is writing a program to display a game's high score list. The line representing the score display has a slope of $-\frac{1}{2}$. What is the slope of a line perpendicular to it?

(A) 2

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

(E) 0

Q3. In a tech company, the slope of the line representing the data usage is 4. What is the equation of a line perpendicular to it that passes through the point (2, 5)?

(A) $y = -\frac{1}{4}x + \frac{13}{2}$

(B) $y = -\frac{1}{4}x + \frac{15}{2}$

(C) $y = \frac{1}{4}x - \frac{5}{2}$

(D) $y = \frac{1}{4}x - \frac{5}{2}$

(E) $y = -\frac{1}{4}x + \frac{13}{2}$

Q4. A gamer is trying to beat a level with a slope of -5 . What is the slope of a line perpendicular to it?

(A) $-\frac{1}{5}$

(B) $\frac{1}{5}$

(C) 5

(D) -5

(E) 0

Q5. In a coding challenge, the equation of a line representing the score is $y = 2x - 3$. What is the equation of a line perpendicular to it that passes through the point (1, 4)?

(A) $y = -\frac{1}{2}x + \frac{9}{2}$

(B) $y = -\frac{1}{2}x + \frac{9}{2}$

(C) $y = \frac{1}{2}x - \frac{9}{2}$

(D) $y = \frac{1}{2}x - \frac{9}{2}$

(E) $y = -\frac{1}{2}x + \frac{9}{2}$

(F) $y = -\frac{1}{2}x + \frac{9}{2}$

(G) $y = \frac{1}{2}x - \frac{9}{2}$

(H) $y = \frac{1}{2}x - \frac{9}{2}$

(I) $y = -\frac{1}{2}x + \frac{9}{2}$

(J) $y = -\frac{1}{2}x + \frac{9}{2}$

(K) $y = \frac{1}{2}x - \frac{9}{2}$

(L) $y = \frac{1}{2}x - \frac{9}{2}$

(M) $y = -\frac{1}{2}x + \frac{9}{2}$

(N) $y = -\frac{1}{2}x + \frac{9}{2}$

Q6. The slope of a line representing a game's progress is -3 . What is the slope of a line perpendicular to it?

(A) $\frac{1}{3}$

(B) $-\frac{1}{3}$

(C) 3

(D) -3

(E) 0

Q7. In a tech firm, the equation of a line representing data is $y = -x + 2$. What is the equation of a line perpendicular to it that passes through the point (3, 1)?

(A) $y = x - 2$

(B) $y = x + 1$

(C) $y = -x - 1$

(D) $y = x - 1$

(E) $y = -x + 1$

Q8. A gaming console's slope of the line representing downloads is 6. What is the slope of a line perpendicular to it?

(A) $-\frac{1}{6}$

(B) $\frac{1}{6}$

(C) 6

(D) -6

(E) 0

Open Ended Questions (FRQ)

Q9. In a coding competition, a line has a slope of 4 and passes through the point (1, 3). Find the equation of a line perpendicular to it that also passes through the point (2, 5), showing all working.

Q10. A gamer wants to find the equation of a line that is perpendicular to the line $y = -2x + 1$ and passes through the point (4, 6). Find the equation of the line.

Chef's Tips

- Tip 1: Emphasize the importance of the negative reciprocal relationship for perpendicular lines.
- Tip 2: Ensure students are applying the concept of slope in various contexts, especially in tech and gaming scenarios.
- Tip 3: Remind students that when writing equations of lines, they must choose the correct form (slope-intercept or point-slope) based on the given information.